Epidemiological Modeling: Spatial SIR (PDE) Model

Project Report Mathematical Modelling; ACLS AS25

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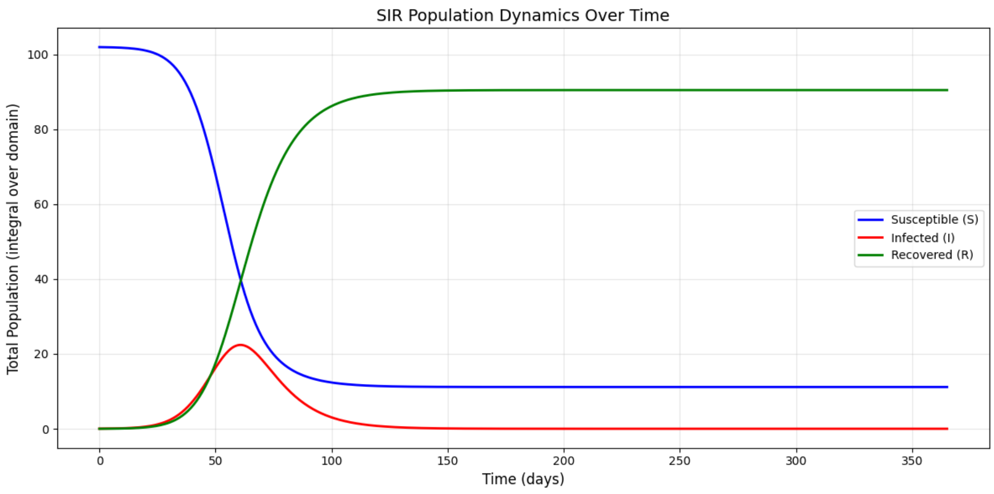

SIR-Model: S, I, R over time

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1 Introduction

In this project we consider a spatial version of the Susceptible-Infectious-Recovered (SIR) model. ... Maybe not needed ?

2 Model Formulation

We consider a spatially distributed population in two dimensions over time and model the spread of an infectious disease. The population density $N(x, y, t)$ is conserved and composed of three compartments: Susceptible (S), Infectious (I), and Recovered (R). The PDE system is given by:

$$\frac{\partial S}{\partial t} = D_s \nabla^2 S - \beta \frac{SI}{N}, \tag{1}$$

$$\frac{\partial I}{\partial t} = D_i \nabla^2 I + \beta \frac{SI}{N} - \gamma I, \tag{2}$$

$$\frac{\partial R}{\partial t} = \gamma I. \tag{3}$$

With the following parameters:

- $\beta > 0$: transmission rate (time^{-1})

- $\gamma > 0$: recovery rate (time^{-1})

- $D_s \geq 0$: diffusion coefficient of susceptible individuals (space^2/time)
- $D_i \geq 0$: diffusion coefficient of infected individuals (space^2/time)

The infection term $\beta \frac{SI}{N}$ transfers individuals from S to I, and the recovery term γI transfers individuals from I to R. The diffusion terms in S and I model random movement of individuals, leading to the spatial spread of the disease.

The spatial SIR model defined by equations 1 to 3 is a time-dependent, non-linear reaction-diffusion system. The laplacian operator ∇^2 in the diffusion terms make the system second-order and the coupling term $\beta \frac{SI}{N}$ makes it non-linear, as the dependent variables S and I are mutliplied.

3 Model Assumptions and Choices

To simulate the system, we make the following assumptions and choices:

- Finite Difference Scheme: We employ the Forward-Time Central-Space (FTCS) scheme. This explicit method simulates the state at t_i directly from the known values at t_{i-1}, while spatial derivatives (∇^2) are approximated using the central finite difference method.
- Boundary Conditions: We assume an isolated population with no movement in or out of the defined space. This is enforced with Neumann boundary conditions (no-flux is implemented by equating boundary values to their neighbouring interior values) to the compartments S and I - R does not diffuse.
- Initial Conditions: The model is initialized with a uniformly distributed susceptible population and a small localized concentration of infected individuals (I) is planted in the center of the grid to simulate the outbreak.
- Parameter Choices: Model parameters are assumed to be constant. Separate diffusion coefficients allow to model different behaviours of Susceptible (S) and Infected (I) individuals. Parameter values are chosen based on the given typical values in the project description and varied to explore their influence on the model.

CITATION LEVEQUE AND https://www.medrxiv.org/content/10.1101/2020.06.24.20139071v2.full

4 Discretisation

Space is discretised on a uniform grid with spacings $h = \Delta x = \Delta y$ and time with step size Δt. The differential operators are approximated as follows:

- Spatial Discretisaton: The Laplacian operator $\nabla^2 u$ in the diffusion term is approximated using a five-point stencil, which is second-order accurate. For any grid point (i,j):

$$\Delta^2 u_{i,j} \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2}$$

- Time Stepping: We use the explicit Forward Euler method to advance the solution in time. Following the general discrete-time formulation $x_{n+1} = f(x_n)$ the state at the next time step is computed directly from the current values:

$$u^{n+1} = u^n + \Delta t \cdot \frac{\partial u^n}{\partial t}$$

Numerical boundary conditions are enforced by the previosuly mentioned Neumann boundary conditions.

5 Stability and Error Analysis

- Step-wise Order of Error: The FTCS scheme combines a forward difference in time ($\mathcal{O}(\Delta t)$) and a central difference in space ($\mathcal{O}(h^2)$). Hence, the local truncation error of the approximation scales like $\mathcal{O}(\Delta t, h^2)$.

- Stability Condition: As an explicit method, the FTCS scheme is only conditionally stable. To ensure numerical stability, the time step Δt is constrained by the diffusion limit (the time step must be sufficiently small relative to the spatial grid size): $\Delta t \leq \frac{1}{2D\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)}$

  This restriction follows from a method of lines stability analysis. Spatial discretisation of the diffusion term leads to a linear ODE system $\dot{\mathbf{u}} = D\,L\,\mathbf{u}$, where L is the discrete Laplacian matrix. For Forward Eulor time integration is stable only if the condition $1 + \Delta t\, D\, \lambda \in [-1, 1]$ is satisfied for all eigenvalues λ of L, whose largest magnited scales as $4(\Delta x^{-2} + \Delta y^{-2})$, yielding the stated restriction on Δt. In our implementation an additional safety factor of 0.5 to this limit.

6 Implementation

The model simulates disease spread across a 100×100 grid using the finite difference method. Rather than treating the population as one large group, it tracks how the disease spreads spatially, as if observing it move across a map. Each cell tracks susceptible (S), infected (I), and recovered (R) populations. Diffusion between neighbouring cells is controlled by coefficients D_s and D_i, with zero-flux Neumann boundaries.

6.1 Initial Conditions Setup

Separate grids were created for each state (S, I and R). The infected population (1% of total) was initialized at the grid center using a Gaussian distribution, with susceptible individuals (99%) uniformly distributed and recovered population initially zero.

6.2 Time stepping routine

The model uses the Euler method, following a step-wise approach. The time step was chosen to ensure numerical stability. The maximum stable time step was calculated as $dt_{\max} = \frac{1}{2 \cdot D_{\max} \cdot (\frac{1}{dx^2} + \frac{1}{dy^2})}$. The actual time step used was $dt = 0.5 \times dt_{\max}$, incorporating a safety factor of 0.5 to prevent the solution from becoming unstable during the simulation.

At each time step implemented in the sir_step function, Neumann boundary conditions were applied to the susceptible and infected populations. The total population was calculated as $N = S + I + R$, and the reaction terms for infection ($\beta \frac{SI}{N}$) and recovery (γI), as well as the diffusion terms ($\nabla^2 S$ and $\nabla^2 I$), were computed using finite differences.

All populations were then updated according to $S^{n+1} = S^n + dt \cdot (D_s \nabla^2 S - \text{infection})$, $I^{n+1} = I^n + dt \cdot (D_i \nabla^2 I + \text{infection} - \text{recovery})$, and $R^{n+1} = R^n + dt \cdot \text{recovery}$. Any negative values were set to zero to ensure physical validity.

6.3 Simulation of the PDE

The simulation was initiated by the `run_simulation` function, which combined all the model's components. This function initialised the spatial grids with the defined initial conditions and set the epidemiological parameters (β and γ) and the diffusion coefficients (D_s and D_i). Four scenarios were simulated over 365 days (~73,000 steps). Snapshots were saved every 50 steps for analysis.

Scenario 1	Scenario 2	Scenario 3	Scenario 4
low β low γ	high β low γ	low β high γ	high β high γ

7 Stability Investigation

dt_max detlat

Grid x 0.15 (lösung nicht mehr stimmen)

8 Parameter Studies

The following plot shows the impact of different transmission (β) and recovery (γ) rates on the spatial dynamics. Each row shows a different scenario. These scenarios are described in the section "Simulation of the PDE". Each column shows a different day. The countdown begins at day 0 and ends at day 365. Each scenario begins with a small number of infected individuals in the middle.

The plot clearly shows that when transmission is low and the recovery rate is high, the number of infected people on the last day was the highest. It can be observed that a high transmission causes the disease to spread faster towards the edges, while a low transmission keeps the infected population for a longer time concentrated in the middle.

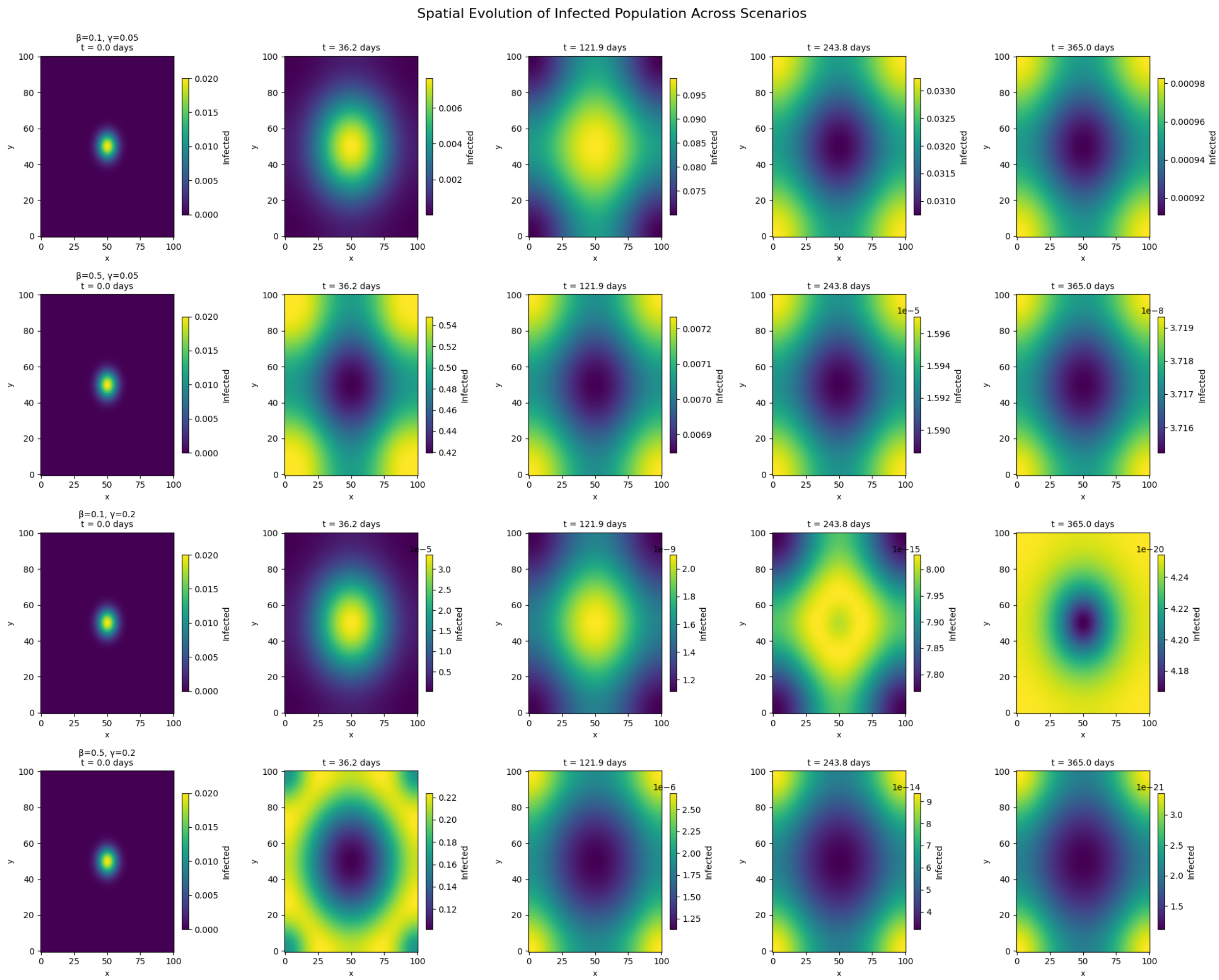

Figure 1: Spatial snapshots at selected time points for all scenarios